As golfers adopt launch monitors, swing sensors, and mobile coaching platforms, demand for accessible, AI-powered form analysis continues to climb. A dedicated app that helps players understand their swing, track progress, and receive personalized feedback fits directly into this expanding market and benefits from the sport's shift toward tech-enabled improvement.

1. **Intro:**

"Formy" is an app (web/native tbd) that analyzes golfers (other sports TBD) swing form using MediaPipe Pose. It provides biomechanical insights, corrective feedback, and progress tracking. The app is designed for recreational and competitive users who want to improve efficiency, reduce injury risk, and monitor progress over time.

- Set an improvement goal
- Take a capture in video
- Insights: core metrics with feedback
- Track improvement vs your goal (progress indicator)

Note: this was originally intended as a multi-sport app focused first on running at launch, however the Ochy app was released before I got this off the ground.

2. **Assumptions:**

- We do not offer medical advice

3. **Monetization:**

- (MVP) Monthly plan to access premium features of the app
- (TBD) Connect with physio clinics for better field data/captures
- (TBD) Gear improvement recommendations / referrals

Pricing model:

- Two tiers:
    - Starter (free) - create profile, make goal, save capture, get basic insights
    - Premium (paid monthly): Set multiple goals, More capture history, Advanced insights (detailed joint angles, stride-phase breakdowns, charts, expanded coaching tips), Video review with live insights (playback with skeleton overlay + metrics), Track and chart/trend improvement vs goals (time-series graphs, comparisons, exportable data)

4. **Features**

User Profile

- Name
- Age
- Gender
- Sports (if expanded beyond golf)
- Goals

Goals

- Type
- Measurement
- Improvement values
- Badges

Captures

- Take a capture
- Or upload existing video
- Historical captures
- Capture analysis

Insights

- Insight information
- Insight details basic
- Advanced insight data (paid)

Admin Panel

- Edit sports
- Edit mediapipe values

5. **Deliverables:**

Screens:

- **Landing screen** - marketing/splash page
  - Full page image of golfer
  - Get started button

- **Login Screen** -
  - Magic link field/button
  - Social sign in

- **Home Screen** - default page, shows goals, insights, and progress
  - Profile header
  - Goals card
    - Pick predefined goal
    - Set values to improve
  - Insights card
    - Show top insight
    - Show improvement
  - CTA upgrade to premium
  - Bottom NAV (Home, Insights, New Capture, History, Profile)

- **Capture Screen -**
  - Record new capture
  - Upload a video

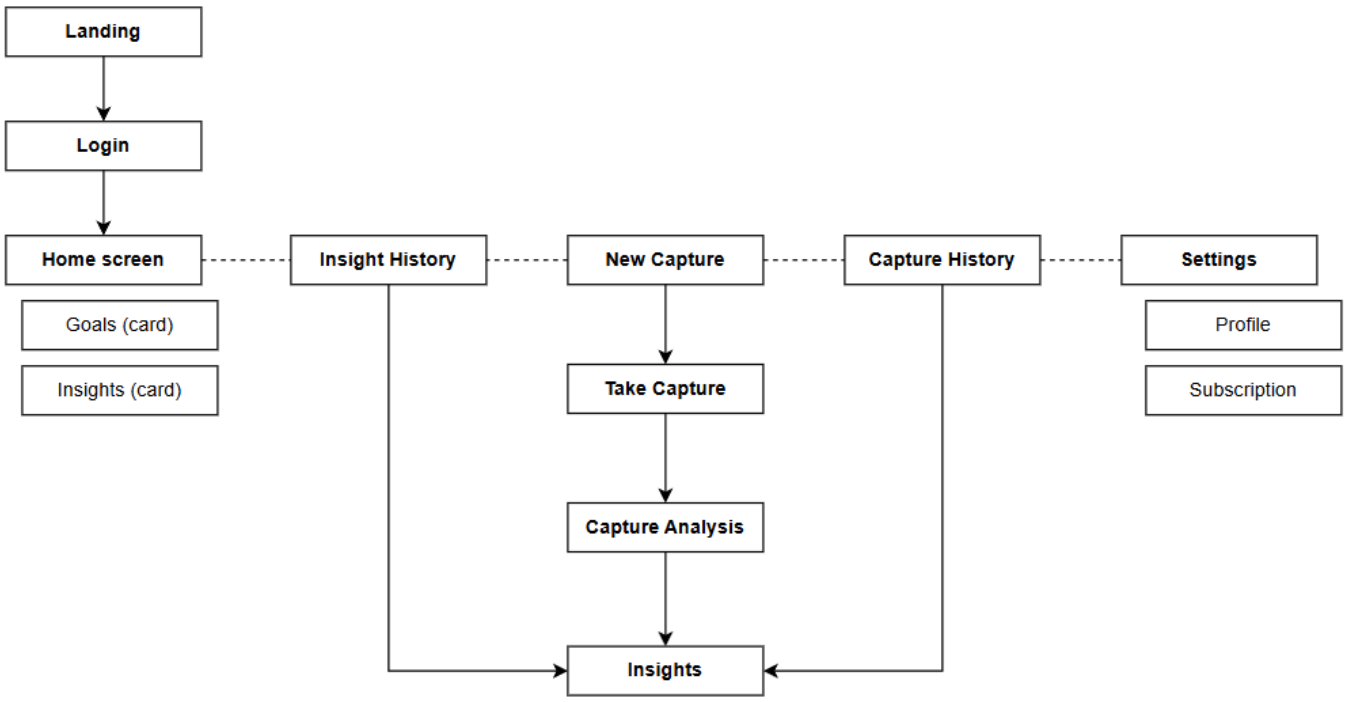
    - Auto progress to analysis screen once capture sufficient
    - Bottom nav

- **Capture Analysis Screen -**
    - Shows capture metadata
    - Shows insights found
    - Card to view each insight text basic / paid
    - Bottom nav

- **History Screen -**
    - Shows historical captures basic / paid
    - Shows insights slider card

- **Admin Screen** - management of sports and mediapipe “good values” in range
    - Desktop app
    - Pages for sports (golf MVP), mediapipe landmarks, insights
        - Tabs for specifics

Navigation:



6. **Stack**

   TBD

7. **Data:**

   user_profiles: full_name, email, age, gender, subscriptionStatus, goals, is_premium, is_admin, is_active

   goals: goalId, goalType, createdAt, captureValues, improvementPercent, is_active

   capture: id, userId, timestamp, videoUrl, landmarks

   insights: id, captured, metricName, value, inRange, feedbackText, is_active

   metric_standard: id, metricName, landmarkIds, landmarkNames, goodRangeLow, goodRangeHigh, lowFeedback, highFeedback, phase, is_active

8. **MediaPipe Pose Metrics Specification**

Metric Name	MediaPipe Pose Landmark IDs	Landmark Names	Good Range	Low Feedback (Below Range)	High Feedback (Above Range)
Spine Angle (Address)	11,23,25 / 12,24,26	Shoulder, Hip, Knee	30–40° forward tilt	“Bend slightly more from your hips to set a stable spine angle.”	“Straighten up a little — you’re bending too far forward.”
Shoulder Tilt (Address)	11,12	L Shoulder, R Shoulder	0–5° (slight right tilt for right-handed golfer)	“Let your trail shoulder drop slightly to set up properly.”	“Level your shoulders more — too much tilt at address.”
Hip Rotation (Backswing)	23,24	L Hip, R Hip	30–45° rotation	“Turn your hips more — backswing is too restricted.”	“Reduce hip turn — you’re over-rotating and losing stability.”
Shoulder Rotation (Backswing)	11,12	L Shoulder, R Shoulder	70–90° rotation	“Turn your shoulders more to create coil and power.”	“Limit your shoulder turn — you’re over-rotating past control.”
Arm Lift (Lead Arm Elevation)	11,13,15 (L)	Lead Shoulder, Elbow, Wrist	60–80° relative to torso	“Lift your lead arm higher — swing is too flat.”	“Lower your lead arm slightly — swing plane is too steep.”

Lead Arm Straightness	11,13,15 (L)	Lead Shoulder, Elbow, Wrist	160–180° (near straight)	“Straighten your lead arm to maintain width.”	“Soften your lead arm slightly — avoid locking it out.”
Trail Elbow Flexion (Top of Backswing)	12,14,16 (R)	Trail Shoulder, Elbow, Wrist	70–100°	“Bend your trail elbow more to load properly.”	“Reduce trail elbow bend — arm is collapsing too much.”
Wrist Hinge (Top of Backswing)	13,15 (L) / 14,16 (R)	Elbow, Wrist	70–90°	“Hinge your wrists more to set the club.”	“Reduce wrist hinge — you’re over-cocking the club.”
Pelvic Lateral Shift (Downswing)	23,24	Hips	2–5 cm toward target	“Shift your hips toward the target earlier.”	“Avoid excessive slide — rotate more instead of sliding.”
Shoulder–Hip Separation (X-Factor)	11,12,23,24	Shoulders, Hips	30–50°	“Increase your shoulder turn relative to hips for more power.”	“Reduce separation — too much torque reduces consistency.”
Lead Knee Flexion (Impact)	23,25,27 (L)	Lead Hip, Knee, Ankle	20–30°	“Flex your lead knee slightly to stabilize impact.”	“Straighten your lead knee a bit — too much bend reduces power.”
Trail Heel Lift (Impact)	28,30,32 (R)	Trail Ankle, Heel, Toe	Heel lifted 2–6 cm	“Allow your trail heel to lift naturally through impact.”	“Keep your trail heel down longer — lifting too early.”
Head Stability (Throughout Swing)	0,7,8	Nose, Ears		“Keep your head steadier — too much movement.”	“Allow a small natural shift — you’re staying too rigid.”
Arm Extension (Follow-Through)	11,13,15 / 12,14,16	Shoulder, Elbow, Wrist	150–180°	“Extend your arms more through impact for better release.”	“Relax your arms — overextending past natural release.”

Torso Rotation (Follow-Through)	11,12,23,24	Shoulders, Hips	90–110°	“Rotate your torso more to complete your follow-through.”	“Avoid over-rotating — maintain balance through finish.”

Angle Calculations

- Joint angle = angle between two vectors
- Rotation = angle between projected vectors on horizontal plane
- Lateral shift = difference in X-coordinates normalized by hip width

Swing Phase Detection

1. Address
2. Takeaway
3. Top of Backswing
4. Downswing
5. Impact
6. Follow-Through

Phase detection can be done via:

- Wrist height
- Shoulder rotation
- Club proxy (wrist–wrist vector)
- Hip rotation velocity

Feedback Trigger Logic

- Flag if ≥30% of frames in a phase exceed thresholds
- Or if 2 of last 3 swings show the same issue

Visibility Requirements

Ignore frames where required landmarks have visibility <0.6.

9. **Mockups:**

formy

Weclome to formy

Start analyzing your form today and get personalized analysis.

+1 Phone number

Send code

or

Apple

Facebook

Google

Strava

Hi, Samuel

Your goals

Add goal +

Increase

Add a goal

Select a goal and measure your improvements over time using form insights.

Form insights are prioritized based on the objective of the goal.

Increase Performance

Reduce Injury Risk

Improve Endurance

Increase Efficiency

Skip

Hi, Samuel

Your goals

Add goal +

Increase Performance

68%

+

New capture

Top insights

All insights →

Keep up the good work!
Your form work has got you 68% improvement in your performance goal

Arms
The user friendly feedback text for the insight goes here

Head and neck
The user friendly feedback text for the insight goes here

Capture

Record a new capture

Hold or secure your phone to get a stable side view.

Start capture

Or upload a video

Capture

Capture analysis

Golf-01
01/02/2025

2 insights

You have two new insights!
Use these to work on your form, then do more captures to track your progress

Arms
The user friendly feedback text for the insight goes here

Head and neck
The user friendly feedback text for the insight goes here

Captures

3 with insights

Capture history

More captures →

Golf-01
01/02/2025
2

Golf-02
01/02/2025
1

Golf-03
01/02/2025
4

Captures

3 with insights

Capture history

More history →

Running

Insights

Running capture

Golf-01
01/02/2025
2 insights

You have two insights!
Use these to work on your form, then do more captures to track your progress

Arms
The user friendly feedback text for the insight goes here

Head and neck
The user friendly feedback text for the insight goes here